Deep Learning Based Medical Image Analysis For Early Disease Detection

Performance Results

Phase III project submission | 30 October 2026

Student / institution: enter the approved cover details. No identity or approval has been invented.

Experiment status

One genuine experiment has completed: ISIC 2016 Task 3 binary benign/malignant classification with ImageNet-pretrained ResNet-18. Seed 42; 738 training, 158 validation, and 379 untouched official test images. Validation loss selected the checkpoint (0.6302 at epoch 4 of 8); the official test partition was held out from model selection. A subset of non-identifying audit and experiment artifacts is included for app reproducibility.

This is an academic research prototype, not a clinically validated diagnostic system. Results must not guide medical care.

Held-out test metrics

Metric	Held-out official test result
Accuracy	0.7810
Macro precision	0.6817
Macro recall / sensitivity	0.7279
Macro F1-score	0.6965
Macro ROC-AUC (OVR)	0.8076

Class-wise outcomes

Class	Precision	Sensitivity	Specificity	F1	Support
benign	0.9018	0.8158	0.6400	0.8566	304
malignant	0.4615	0.6400	0.8158	0.5363	75

Confusion matrix

Class order benign, malignant. Matrix: [[248, 56], [27, 48]] (rows are reference labels; columns are predicted labels).

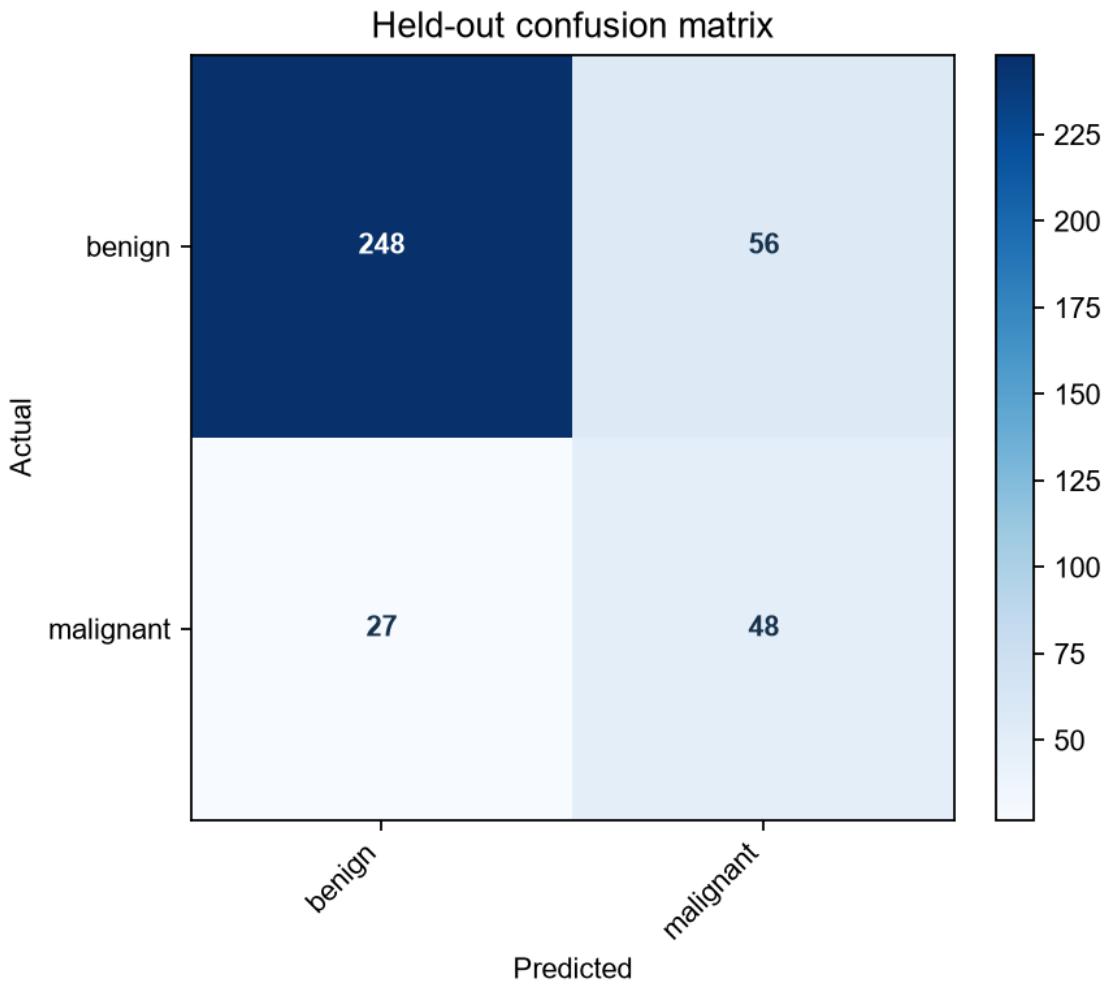


Figure 1. Confusion matrix computed from the official test set.

Training and validation



Figure 2. Per-epoch training history; model selection used validation loss.

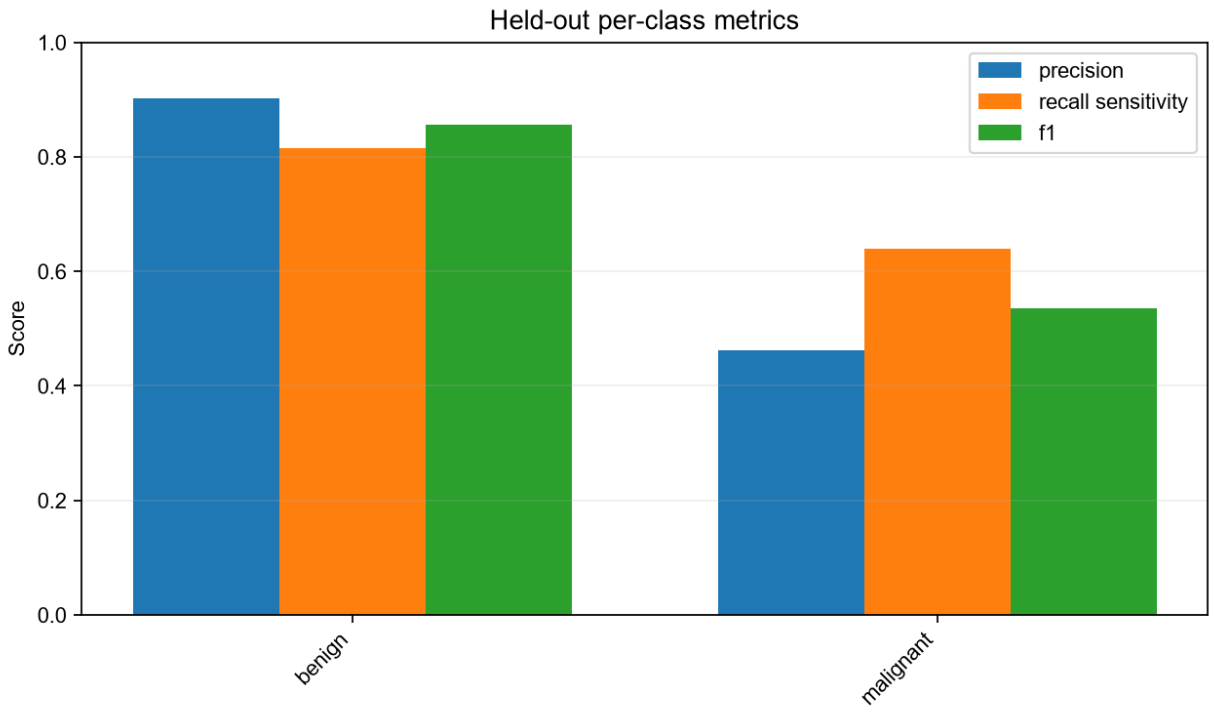


Figure 3. Class-wise held-out test metrics.

Best validation loss: 0.6302144312405888. Training epochs: 8. Checkpoint selection did not use test metrics.

Interpretation and model choice

ResNet-18 is the only trained candidate, so no comparative winner is claimed. Overall accuracy can obscure minority malignant errors: malignant precision and F1 are materially lower than benign performance, and the test set contains fewer malignant examples. The checkpoint is the selected model only for this run, chosen by validation loss; the measured scores do not demonstrate clinical utility.

NIH ChestX-ray14 is not locally available because remaining workspace storage is insufficient for its full release alongside current files. APTOS 2019 requires Kaggle account sign-in and acceptance of competition rules; access is pending user download. Thus these two tasks are implemented in code but not trained/evaluated.